

Introduction to Research Data Management (RDM)

Session Objectives

By the end of this session, you'll be able to:

- Define research data, research data management, and the FAIR, CARE, and OCAP principles.
- List the different stages of the research data management lifecycle.
- Articulate how RDM principles support research transparency, accessibility and reproducibility.
- Identify where to find RDM support.
- Define sensitive data.
- Discuss risk assessment and harm in data management at both the individual participant and global level of privacy and data security.
- Connect RDM concepts and practices to previous research and/or data management experiences.

What is Research Data Management (RDM)?

What are research data?

Every research project seeks to investigate a research question. The materials that support the answering of a research question are considered data. This can include:

- Spreadsheet data
- Audio recordings
- Video recordings
- Textual transcripts
- Images / photographs
- Corpora of text
- Computer code
- Marginalia (writing in the margins of books)
- Geographic information, such as coordinates
- Others?

What data are you working with?

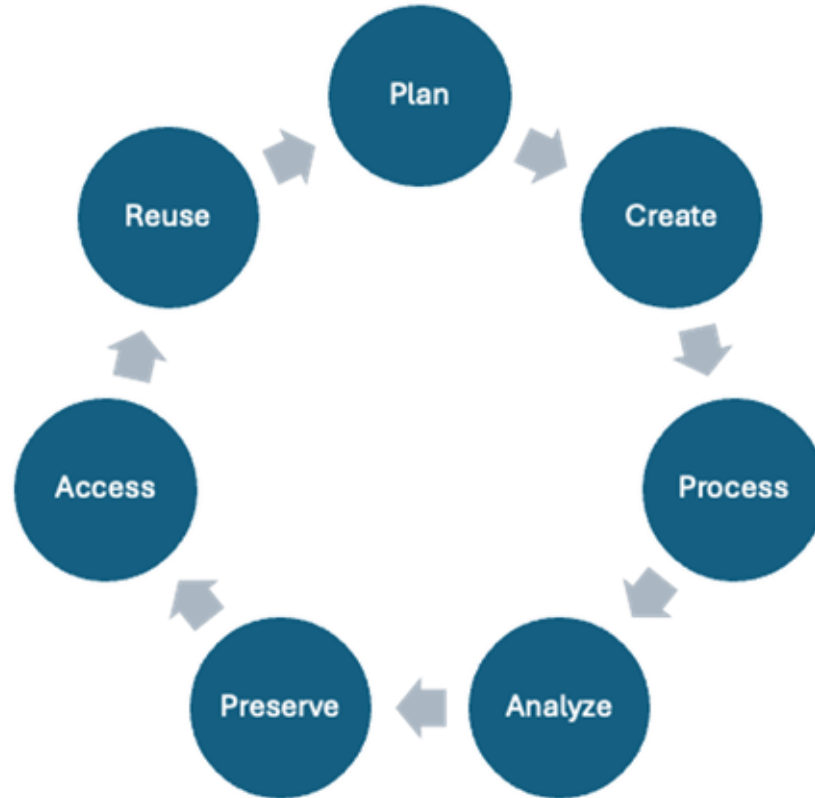
Share your answers in the chat, or raise a hand!

What is RDM?

RDM involves the practices and activities over the research lifecycle that pertain to the management of research data, including planning, collecting, analyzing, documenting, storing, sharing, and preserving.

Ultimately, RDM is research.

The research data lifecycle



Why Should You Care?

What does academic integrity mean to you?

Share your answers in the chat, or raise a hand!

Academic integrity

- For undergraduates, academic integrity revolves around plagiarism (cite your sources!)
- As you move into graduate research, there is a shift from 'you as consumer' of information to 'you as producer' of information.
- From this perspective, academic integrity encompasses the integrity of your research data, as your data forms the basis of the claims you will make in your research papers.

Consider the following

Stanford president resigns after fallout from falsified data in his research

Updated July 20, 2023 at 5:56 PM ET | 6

The Associated Press



People walk on the Stanford University campus beneath Hoover Tower in Stanford, Calif., on March 14, 2018. Stanford President Marc Tessier-Laudan said on Wednesday he would resign, citing an independent review that showed him at research misconduct had found false in other papers authored by his lab.

Research Fraud: Is Everything We Think We Know About Alzheimer's Disease Wrong?

TOPICS: Alzheimer's Brain Dementia Popular University Of Michigan

By MICHAEL MCKENZIE, UNIVERSITY OF MICHIGAN AUGUST 23, 2018



Evidence revealed that scientists had faked images in a study published at years ago. This was an important paper that other researchers had trusted and relied on as they did their own work.

University investigation found prominent spider biologist fabricated, falsified data

Co-authors of now-retracted papers from Jonathan Pruitt were sent summarized findings that the scientist violated McMaster University's research integrity policy

[source](#)

[source](#)

[source](#)

Consider the following

- In each of these stories, purposeful academic fraud occurred because the data was undocumented and inaccessible.
- While these examples represent extreme cases, what happens more commonly is that poor data management practices result in ***accidental misrepresentations of data.***

Can you think of other reasons that data might be misrepresented?

Share your answers in the chat, or raise a hand!

Why should you care?

Our goal is to teach you how you can avoid accidental misrepresentation by focusing on the integrity of your data, and how to incorporate each stage of the research data lifecycle into your overall program of research.

Data Policies

Have you encountered any data policies?

- Publishers
- Funders
- Your institution
- Your supervisor

Share your answers in the chat, or raise a hand!

Publisher policies

It is becoming more common for publishers to require that research data be shared upon publication to support the transparency, reproducibility, and validation of the research findings.

Some examples include:

- Springer Nature: [Research Data Policy](#)
- Wiley: [Data Sharing Policies](#)
- PLOSOne: [Data Availability](#)

Funder policies

Research funders are also starting to create data policies.

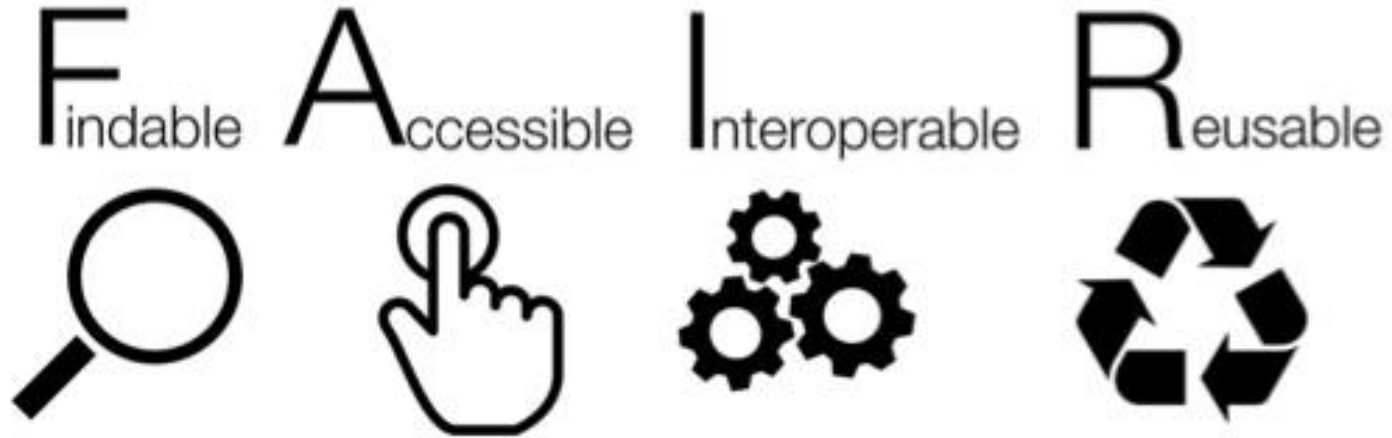
In Canada, the Tri-Agency RDM Policy, has outlined the following requirements:

- Data Management Plans
- Data Deposits
- Institutional Strategies

Why do you think these policies are being created?

Share your answers in the chat, or raise a hand!

FAIR Principles



FAIR Principles

FAIR does not mean that all data must be open or public.

Rather, FAIR supports the structured discoverability and usability of data and other research outputs.

CARE Principles



Principles of OCAP



Sensitive Data

What types of materials would you be hesitant to post publicly?

Share your answers in the chat, or raise a hand!

Sensitive data

The Sensitive Data Toolkit for Researchers defines sensitive data as “information that must be safeguarded against unwarranted access or disclosure”. Examples include:

- Personal information
- Personal health information
- Educational records
- Customer records
- Financial information
- Criminal information
- Geographic information (e.g., detailed locations of endangered species)
- Information that is deemed to be confidential
- Cultural or community information/data
- Proprietary information
- Information that is protected by data sharing agreements or other contracts

Sensitive data

The Sensitive Data Toolkit for Researchers provides a framework for assessing and protecting participant risk, but it is good practice to ask yourself the following questions:

- Can someone be identified?
- Could harm result from sharing?
- What consent was given?
- Are there legal restrictions?
- Are there community expectations?

Advisory

Ethically managing sensitive data is beyond the scope of this course, and we advise you to consult with your institutional supports to determine adequate strategies for addressing sensitive data in your research.

Data Storage

Data storage

- Where do your research materials live right now?
- Have you ever accidentally lost or overwritten your work?
- If your personal computer crashed or was stolen, would you still have access to your work?

Share your answers in the chat, or raise a hand!

Data storage

- Where your data is stored is a key part of data management.
- It is always recommended that your research materials are stored **in at least two places**, and that one of those places is remote cloud storage beyond your personal machine.
- This is known as a back-up.

Data Preservation

In addition to storing data for your active work, there are platforms that support the long-term storage and preservation of your data and research materials after the project's completion.

Storage	Supports active research work	laptop, shared drive, cloud storage
Repository	Publishes and shares research outputs	Borealis, Zenodo
Archive	Ensures long-term preservation	institutional archives, digital preservation systems

Data Preservation

In addition to storing data for your active work, there are platforms that support the long-term storage and preservation of your data and research materials after the project's completion.

Data Preservation

We will talk more about data preservation later in this course, but one of our main goals is to show you how to conduct research in such a way that data, code, and all accompanying materials are able to be shared and interpreted by others after the project is complete.

Breakout!